Partial Fractions with Squared Linear Denominators

Exam code: 9MA0

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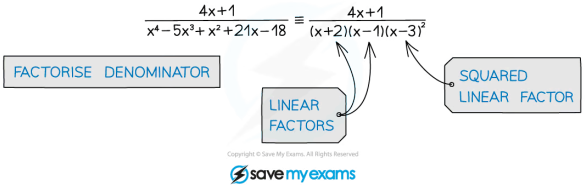
Partial fractions with squared linear denominators

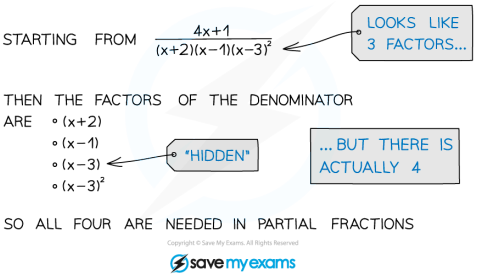
What are partial fractions with squared linear denominators?



- In partial fractions the common denominator is split into parts (factors)
- This is the reverse process to **adding** (or subtracting) fractions
- In harder questions there is a repeated factor, this is a **squared linear factor**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- A **linear factor** is of the form (**ax + b**)
- It is possible **b = 0** so a linear factor could be of the form **ax** (eg **4x**)
- A **squared linear factor** is of the form $(\mathbf{ax + b})^2$
- With b = 0 this would be of the form $(\mathbf{ax})^2$ ($\mathbf{x}^2$ would be too!)



How do I find partial fractions with squared linear denominators?

STEP 1 **Factorise** the denominator (Sometimes the numerator can be factorised too)

STEP 2 **Split** the fraction into a **sum** with a **squared linear denominator** and any other single linear denominators

STEP 3 Multiply by the denominator to get rid of fractions

STEP 4 Substitute values of **x** to find **A**, **B**, etc (or use comparing coefficients)

STEP 5 Write the **original** as partial fractions

MAY INVOLVE
- FACTOR THEOREM
- ALGEBRAIC DIVISION

STEP 2: SQUARED LINEAR DENOMINATOR

$$\equiv \frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{x-3} + \frac{D}{(x-3)^2}$$

STEP 2: OTHER LINEAR DENOMINATORS

STEP 3: MULTIPLY BY DENOMINATOR

$$4x+1 = A(x-1)(x-3)^2 + B(x+2)(x-3)^2 + C(x+2)(x-1)(x-3) + D(x+2)(x-1)$$

STEP 4: CHOOSE VALUES OF x

LET $x=1$: $5=12B$ $\quad B=\frac{5}{12}$

LET $x=3$: $13=10D$ $\quad D=\frac{13}{10}$

LET $x=-2$: $-7=-75A$ $\quad A=\frac{7}{75}$

LET $x=0$: $1=-9A+18B+6C-2D$

$$1=\frac{-21}{25}+\frac{15}{2}+6C-\frac{13}{5}$$

$$6C=\frac{-153}{50}$$

$$C=\frac{-51}{100}$$

NO VALUE OF x GIVES AN EQUATION IN C ONLY SO CHOOSE AN EASY NUMBER TO WORK WITH

STEP 5: WRITE AS PARTIAL FRACTIONS

$$\frac{4x+1}{x^4-5x^3+x^2+21x-18} \equiv \frac{7}{75(x+2)} + \frac{5}{12(x-1)} - \frac{51}{100(x-3)} + \frac{13}{10(x-3)^2}$$

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Worked Example

STEP 1: NUMERATOR CAN'T BE FACTORISED, DENOMINATOR ALREADY IS

$$\frac{x^2-9x+38}{(x+1)(x-3)^2} \equiv \frac{A}{x+1} + \frac{B}{x-3} + \frac{C}{(x-3)^2}$$

STEP 2: THERE IS A SQUARED LINEAR DENOMINATOR

STEP 3: GET RID OF FRACTIONS!

$$x^2-9x+38 \equiv A(x-3)^2 + B(x+1)(x-3) + C(x+1)$$

STEP 4: CHOOSE SUITABLE VALUES OF x

LET $x=3$: $20=4C$ $\therefore C=5$

LET $x=-1$: $48=16A$ $\therefore A=3$

NOTICE WE CAN NOT GET AN EQUATION IN B ONLY BY CHOOSING A VALUE OF x

LET $x=0$: $38=9A-3B+C$

$3B=27+5-38$

$\therefore B=-2$

STEP 5: WRITE ORIGINAL FRACTION AS PARTIAL FRACTIONS

$$\therefore \frac{x^2-9x+38}{(x+1)(x-3)^2} \equiv \frac{3}{x+1} - \frac{2}{x-3} + \frac{5}{(x-3)^2}$$

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